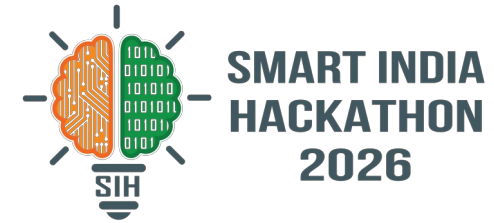
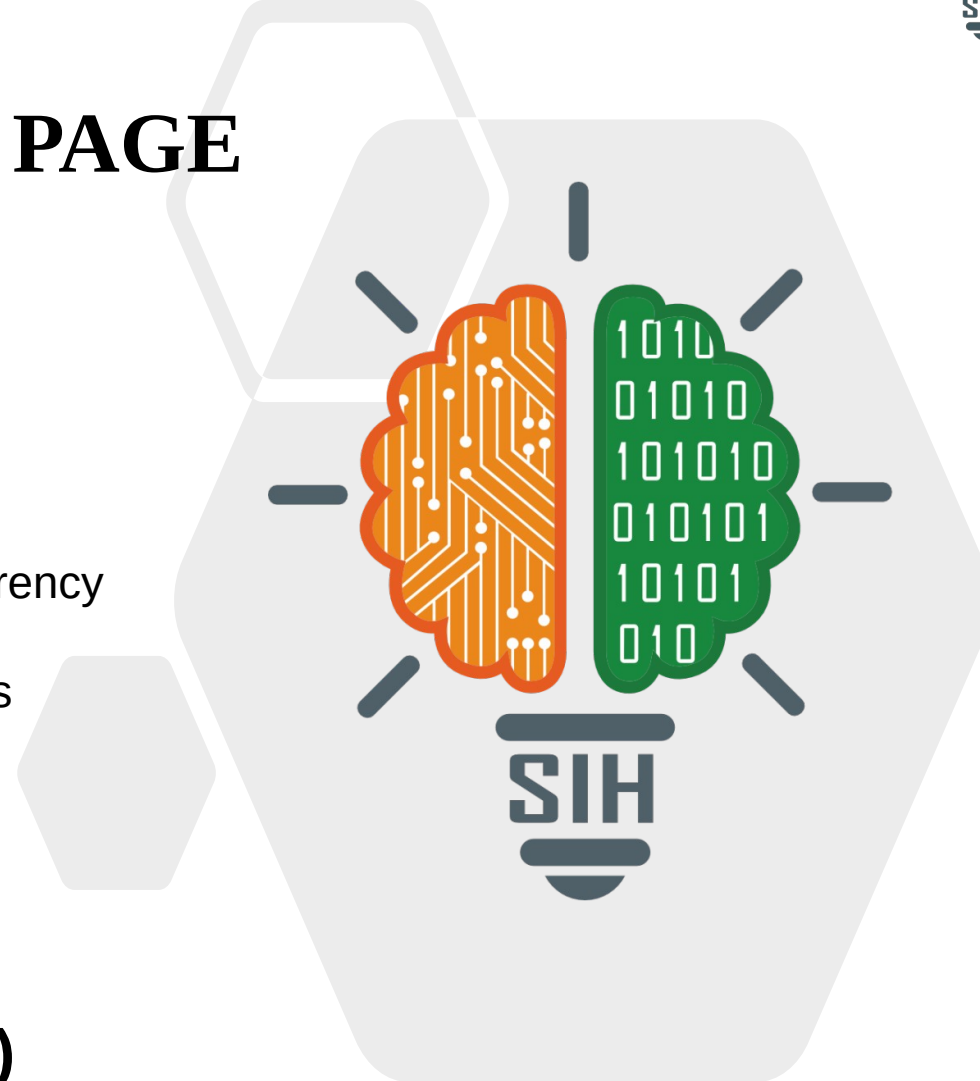


SMART INDIA HACKATHON 2026



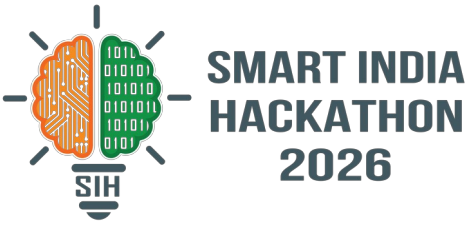
TITLE PAGE

- **Problem Statement ID** – SIH26183
- **Problem Statement Title-**
Real-Time Identification of Fraud-Linked Cryptocurrency
Exchanges from Victim-Reported Suspect Wallet
Addresses through Automated Blockchain Analytics
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 156766
- **Team Name (Registered on portal)**
COLD COFFEE



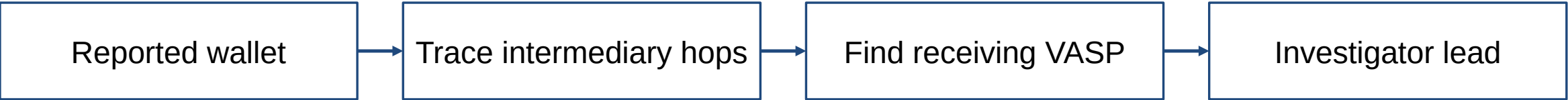


AEGIS FORENSICS



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution

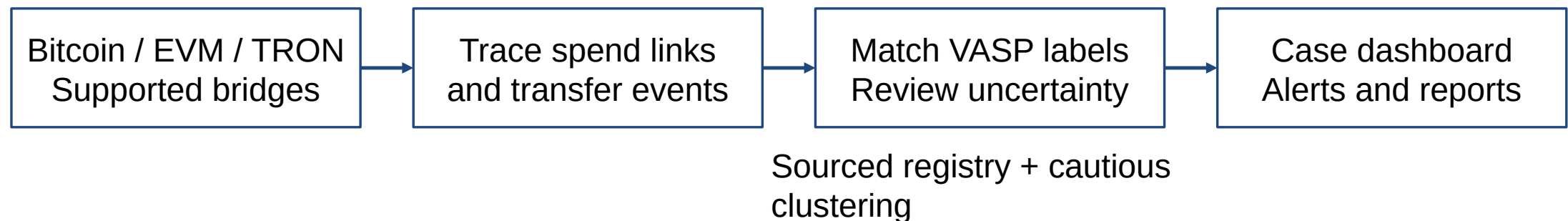


- How it addresses the problem
Movement alerts and sourced fund trails will support investigator review and VASP preservation requests.
- Innovation and uniqueness of the solution
Link each lead to its evidence and next action. Preserve earlier findings as transactions and labels change.

- Technologies to be used (e.g. programming languages, frameworks, hardware)

React, Python/FastAPI, queued workers, incremental blockchain indexes and versioned VASP labels.

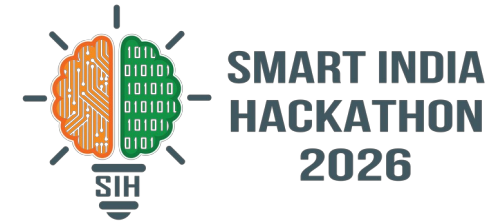
- Methodology and process for implementation (Flow Charts/Images/ working prototype)



Explainable rules and evaluated ML will flag suspect patterns. Mixing and unsupported bridges remain explicit gaps.



FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea

Validate Bitcoin attribution first, then monitoring, additional chains and approved LEA integration.

Label supply: reviewed disclosures and authorized VASP data, retaining source and review history.

- Potential challenges and risks

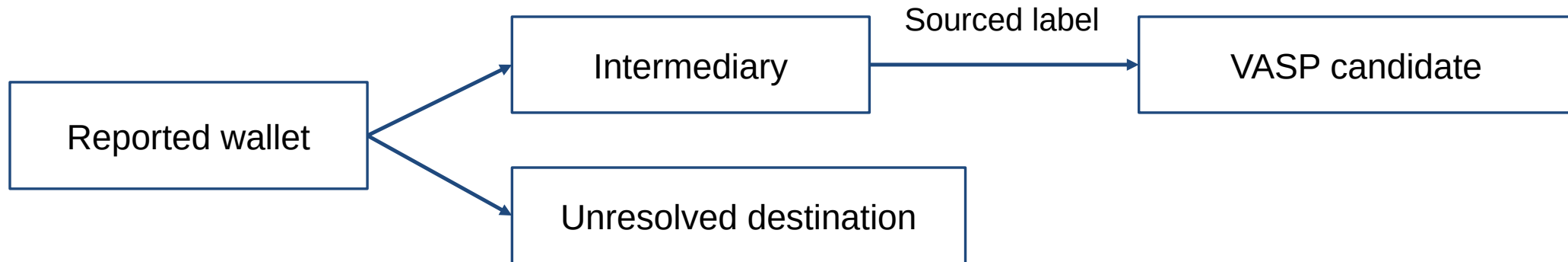
Incomplete labels, ambiguous flows and provider limits can reduce coverage or cause false attribution. Agency data and APIs require authorization.

- Strategies for overcoming these challenges

Index and cache data. Version labels and show unresolved paths. Protect cases with access controls and audit logs. Test held-out paths for precision, coverage and alert latency.

- Potential impact on the target audience

Investigators will receive prioritized VASP leads with reviewable transaction paths and attribution sources.



Fictional illustration. A path does not prove exact fund continuity or wrongdoing.

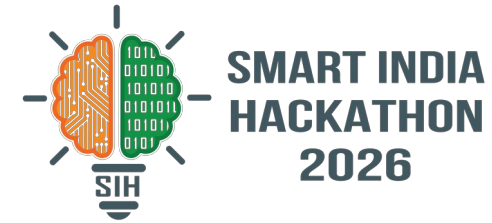
- Benefits of the solution (social, economic, environmental, etc.)

Social: support timely intervention. Economic: reduce repeated tracing work.

Operational: standardize reports and VASP coordination while retaining human review.



RESEARCH AND REFERENCES



- Details / Links of the reference and research work

Address clustering: Meiklejohn et al. (2013), A Fistful of Bitcoins.

[ACM IMC paper](#)

ML-assisted analysis: Weber et al. (2019), Anti-Money Laundering in Bitcoin.

[Research paper](#)

Blockchain data: Blockstream Esplora address and transaction API.

[Official documentation](#)

Attribution: Binance wallet disclosure (2022). Historical, not a deposit registry.

[Published disclosure](#)

AML context: FATF (2021), guidance for virtual assets and VASPs.

[FATF guidance](#)

Evidence handling: NIST SP 800-86. No guarantee of legal admissibility.

[NIST guidance](#)